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Simplifying Scientific Text

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1 Scientific Problem and Project Objectives

1.1 Scientific problem

Scientific and biomedical texts contain specialized terminology, complex sentence structures, numerical findings, and expressions of uncertainty. These features can make reliable research difficult for non-specialist readers to understand. Scientific text simplification aims to improve accessibility while preserving the information expressed in the source. The CLEF SimpleText task provides a shared setting for investigating this problem. [1]

The central difficulty is that a sentence may become shorter and more fluent while losing an important fact, changing negation, reversing an effect, or strengthening an uncertain conclusion. Readability and meaning preservation therefore need to remain distinct considerations when interpreting simplification quality. Our previous planning-based system, SPU_DeepSimplify, addresses this problem through terminology support and controlled generation. [2]

1.2 Project objectives

The project aims to extend SPU_DeepSimplify into an ensemble system that uses outputs from a PEFT-adapted simplification branch and the existing planning-based branch. The final component receives the original sentence and both candidates, then either selects a candidate or generates a new simplification grounded in the source. The purpose is to investigate whether the two branches provide useful complementary outputs. [2]

The published previous system, with SARI 45.00, is the reference system. The project target is SARI of at least 47.00 under a clearly specified evaluation setting. This target is an objective, not an achieved result. Any direct comparison requires the same inputs, references, and scoring procedure. Improvements to the existing branch may also be considered and compared with its published configuration. [2]

1.3 State of the art

The selected SimpleText literature includes strong prompting-based systems, adapted generators, explicit planning, terminology guidance, and multi-candidate processing. Writerslogic reports sentence SARI 47.43 using Claude Sonnet 4 with multiple candidates and heuristic selection. FOSU-YZH reports 47.01 with the original reference setting using statistics-aware prompting and conservative selection. These figures describe their reported configurations and are not a harmonized ranking across every evaluation condition. [3, 4]

AIIRLab combines outputs from adapted and prompted systems using a larger language model. Its ensemble reaches SARI 46.24 with original references, but has a higher FKGL than its DPO component. This result illustrates that a stronger SARI score need not correspond to the lowest estimated reading grade. The current project builds on this motivation while extending our own previous system; it does not claim that PEFT or ensemble simplification is new in itself. [2, 5]

2 Theoretical Background

2.1.1 Text simplification and language generation

Text simplification rewrites complex content in more accessible language. It may involve lexical substitution, short explanations, rephrasing, splitting, or removal of secondary material. The appropriate transformation depends on the source. In sentence-level simplification, one input sentence can produce more than one simplified sentence when splitting improves clarity. Important entities, findings, comparisons, negation, and uncertainty must remain faithful to the source. [1, 2]

2.1.2 Transformer architecture and causal attention

The Transformer uses attention to build context-dependent token representations. Decoder-based language models generate a response autoregressively: each target token is predicted from the preceding prompt and preceding target tokens. Causal masking prevents access to later target tokens. It does not prevent a target token from using source information already present in the prompt. [6]

2.1.3 Supervised fine tuning

Supervised fine-tuning uses examples containing an input and an expected response. For simplification, the complex sentence is the input and the aligned simplification is the supervised response. During training, the response is included so that its tokens can be predicted. With response-only loss, prompt tokens are excluded from the loss calculation. At inference time, the reference response is unavailable and the system generates its own output. [7, 8]

2.1.4 Parameter efficient fine tuning and LoRA

Parameter-efficient fine-tuning adapts a pretrained model while updating a limited set of parameters. LoRA freezes pretrained weight matrices and learns low-rank updates through additional trainable matrices. This reduces the number of trainable parameters compared with full fine-tuning. It supports adaptation with fewer trainable weights, but does not guarantee better generalization or a particular simplification score. [7]

2.1.5 Quantization and QLoRA

Quantization stores weights at reduced numerical precision to lower memory requirements. QLoRA combines a frozen four-bit quantized base model with trainable LoRA adapters. Its primary relevance is memory-efficient adaptation of large language models. Lower memory requirements should be distinguished from training speed and output quality, which depend on the actual setup and require measurement. [8]

2.1.6 Prompting and semantic planning

Prompting guides a model through instructions and examples without necessarily updating its weights. Semantic planning separates sentence analysis from generation. A planner can expose properties such as terminology, structural complexity, and statistical content in a structured representation. A control layer then uses the plan to select an operation before the executor generates the output. [2]

2.1.7 Terminology support and ensemble processing

Terminology support supplies definitions or plain-language hints for difficult source terms. It can assist lexical rewriting, but is not an independent verification of all facts in the output. Ensemble processing combines outputs from several generation paths. Candidate selection returns an existing candidate unchanged; generative fusion produces new wording from the source and candidates. These formulations should be distinguished because fusion can introduce changes beyond the original alternatives. [2, 3, 5]

3 Previous System and CLEF Participation

3.1 Task and published paper

Our team, SPU_DeepSimplify from Syrian Private University, participated in CLEF 2026 SimpleText Task 1, focusing on Task 1.1 sentence-level scientific text simplification. The system is described in the paper SPU_DeepSimplify at the CLEF 2026 SimpleText Track A Planning-Based Approach to Scientific Text Simplification by Raneem Rabih, Riad Sonbol, and Wesam Alsohle. The published system uses pretrained models in inference mode rather than task-specific weight adaptation. [2]

3.2 Architecture and models at each stage

The pipeline begins with MedSimplify terminology support. Relevant source terms are detected and short definitions or safe plain-language hints are prepared. Llama 3.1 8B then acts as the semantic planner and produces a structured JSON analysis. When available, the true next sentence is provided to the planner as local context. [2]

A rule-based strategy layer selects rephrasing, splitting and simplification, omission, or keeping the sentence unchanged. Llama 3.3 70B acts as the executor. It performs terminology-aware processing when applicable and generates the output according to the selected strategy. Both named models were accessed through the OpenRouter API without training updates. [2]

Output validation applies operational checks and fallback behavior. Selective post-processing then revisits selected outputs that remain identical or very similar to the source. The paper does not separately name a model for that post-processing step. The earlier semester report documents the local development pipeline and its interface; the CLEF paper is the source for the published two-model configuration and Codabench result. [2, 9]

3.3 Codabench result

The system processed the supplied official source file and its predictions were evaluated on Codabench. The paper reports SARI 44.91 for the main pipeline and 45.00 after selective post-processing. These are documented previous results, not results of the proposed PEFT ensemble. The input-file size alone does not identify the exact subset or reference setting underlying a platform score. [1, 2]

4 Related Work

4.1 Prompting and Candidate Selection

Writerslogic presented a scientific text simplification system based on generating multiple candidates and selecting one using heuristic rules. The system uses few-shot examples to guide the simplification style and generates five candidates with Claude Sonnet 4 at different temperatures. Candidates are evaluated according to source-word retention, compression, lexical simplicity, and plain-language vocabulary, with a penalty for exact copying. The system achieved a sentence-level SARI score of 47.43. For document simplification, it used Llama 3.3 70B directly and achieved 43.96. The study demonstrates the potential of candidate generation and selection without additional training. However, its selection criteria rely on surface features that cannot independently verify meaning preservation [3].

FOSU-YZH combined statistics-aware prompting with conservative candidate selection. The system generates four candidates using different DeepSeek configurations, including a default version, a version using local context, and a cautiously worded version intended to preserve uncertainty. Unsuitable outputs are removed, and the remaining candidates are evaluated using features such as compression, avoidance of copying, and simpler wording. The default candidate is replaced only when another candidate exceeds it by a predefined margin. The system achieved sentence-level SARI scores of 47.01 with the original references and 46.23 with the automatic references. The study highlights the importance of handling statistical expressions and cautious language. However, candidate selection does not include independent semantic verification, and the complete four-candidate pipeline was used for the main English portion of the data [4].

DUTH used Gemma-2-9B-IT with source-grounded instructions emphasizing the preservation of numerical findings, comparisons, and uncertainty. Sentence-level instructions encouraged simple vocabulary and splitting where appropriate. For documents, the study examined direct simplification and reconstruction from simplified sentences in their original order. Processing included output-length checks, fallback to an alternative when necessary, and light cleaning of repetition and vocabulary. The system achieved SARI 46.09 for sentences and 46.16 for documents. The study demonstrates that prompting and post-processing can produce competitive results without fine-tuning the simplifier. However, instructions and length checks do not independently verify the accuracy of the generated content [10].

FOSU_cr used domain-matched few-shot examples to guide DeepSeek-V3, followed by regular-expression-based post-processing. The system selected three real Cochrane examples according to source and summary length requirements and included them in instructions defining the desired plain-language style. After generation, statistical expressions were removed, and empty brackets and spacing were cleaned. The system reported a local average sentence-level SARI score of 43.50. Its official document-level scores were 48.85 with the original references and 47.97 with the automatic references. The study illustrates the influence of domain-appropriate examples. However, removing statistical expressions and evidence-certainty descriptions may discard important distinctions. The source of the selected examples must also be considered when interpreting the results [11].

4.2 Fine-Tuning and Controlled Adaptation

ELiRF-UPV investigated separating simplification-action prediction from text generation. The system used RoBERTa-large with a Conditional Random Field (CRF) layer to predict operations such as copying, deletion, rephrasing, splitting, and merging within document context. It then examined action execution using prompted Gemma 4 31B or adapted versions of BART and FLAN-T5, including experiments with Direct Preference Optimization (DPO). The best official results were SARI 46.36 for sentences and 46.22 for documents, obtained using Gemma 4 31B without action-plan conditioning. The study shows that separating planning from generation enables examination of the individual components, but does not guarantee better performance. The planner's effect varied, and some locally trained systems did not achieve comparable performance in the official evaluation [12].

NIL-UCM compared zero-shot prompting, few-shot prompting, action classification followed by simplification, LoRA adaptation, and multi-agent processing. Sentence-level experiments used models including Llama 3.1 8B, Mistral 7B, and Gemma 2. The best sentence result was obtained with role-based zero-shot prompting using Llama, achieving SARI 45.51. The LoRA experiments addressed document simplification using 849 abstract–plain-language-summary pairs. The study also examined a multi-agent system that generates four drafts, selects one, reviews factual content and readability, and revises the text. This system achieved document-level SARI 45.34. The study provides a comparison of several approaches, although the tested LoRA configurations did not produce the intended improvement. Moreover, using different models within the multi-agent system makes it difficult to separate the effect of the architecture from the capabilities of its models [13].

Taming CATS: Controllable Automatic Text Simplification through Instruction Fine-Tuning with Control Tokens studied the use of control tokens to specify desired readability levels or length ratios. Experiments covered models from the Llama, Mistral, and Qwen families, ranging from 1 to 14 billion parameters, with LoRA used for models larger than four billion parameters. The approach was evaluated on biomedical, administrative, encyclopedic, and news datasets, including Med-EASi, SimPA, WikiLarge, and Newsela. One biomedical configuration achieved SARI 52.18 using Mistral 7B with Dale–Chall readability control. Qwen3 1.7B achieved SARI 52.09 with character-length-ratio control on Med-EASi. These scores belong to the study's own data splits and configurations. Improvements in SARI did not always correspond to improvements in LENS, highlighting the need to evaluate simplification quality alongside compliance with the requested control attribute [14].

4.3 Planning and Structural Simplification

Plan-Guided Text Simplification with Extended Contexts examined the effect of increasing context length in action-guided simplification. The study compared fine-tuned BART-base with prompted models from the Qwen2.5 and Gemma families, using inputs containing one to five paragraphs from Newsela-auto. It considered reference action plans extracted from alignment, as well as plans predicted by an existing planner. The best result was SARI 51.5, obtained using Qwen 32B with a reference action plan and a single paragraph. This result therefore depends on access to a reference plan and does not represent a fully predicted-plan setting. The study highlights the distinction between providing additional context and successfully following a plan. Its findings concern news data and evaluation settings that differ from biomedical simplification in SimpleText [15].

Let's Simplify Step by Step: Guiding LLM Towards Multilingual Unsupervised Proficiency-Controlled Sentence Simplification proposed gradual simplification toward a specified language-proficiency level. The approach plans transitions between CEFR levels, selects semantically similar examples, and uses conversation history during stepwise generation. The study examined Llama-3-8B, Gemma-2-9B, and Mistral-7B on data covering five languages. It evaluated target-level attainment through exact and adjacent-level accuracy, together with semantic-similarity measures. The complete Llama-based system achieved 37.37% exact accuracy and an STS score of 69.90%. These values describe proficiency-level control and semantic similarity rather than SARI. The study demonstrates a formulation of simplification as a sequence of transitions toward a specified language level [16].

Our previous system, **SPU_DeepSimplify**, introduced a planning-based approach to scientific sentence simplification. It first identifies terminology and obtains plain-language hints from MedSimplify. Llama 3.1 8B then analyzes the sentence and produces a structured JSON plan. A rule-based layer selects an operation, such as rephrasing, splitting, deletion, or preservation, which is executed by Llama 3.3 70B. Output-validity checks and selective post-processing are subsequently applied to outputs that are identical or very similar to the source. Both models were used for inference without fine-tuning. The system achieved a final sentence-level SARI score of 45.00. Its main strength is the traceability of its processing stages. Its limitations include the possibility of planning errors affecting generation and the absence of experiments independently isolating the contributions of all components [2].

4.4 Terminology-Supported Simplification

JEBS: A Fine-grained Biomedical Lexical Simplification Task examined biomedical lexical simplification by separating term identification, replacement-operation classification, and replacement generation. The operations included substitution, explanation, generalization, exemplification, and omission. The study used PLABA data containing 400 abstracts, 10,314 term occurrences, and 21,595 manually annotated simplifications. It examined detection and classification models from the BERT and DeBERTa families, alongside generation models such as Llama and GPT-4o-mini. Detection and classification were evaluated using F1, while generated replacements were assessed by human evaluators for simplicity, accuracy, completeness, and conciseness. This decomposition supports detailed analysis of terminology handling. However, its results do not constitute an evaluation of a complete sentence simplification system, and no overall system-level SARI score was reported [17].

Medical Text Simplification: From Jargon Detection to Jargon-Aware Prompting investigated whether identifying difficult terminology and including it in prompts could improve medical text simplification. The study used models such as BERT, RoBERTa, BioBERT, and PubMedBERT for jargon detection, followed by simplification with Llama 3.1 8B and Medicine-Llama3-8B under different prompting configurations. The simplification evaluation included 300 PLABA abstracts containing 3,315 sentences. Llama with reference operations achieved SARI 32.26 and FKGL 15.36, compared with SARI 29.87 and FKGL 13.53 under direct prompting. These results show that an improvement in SARI can accompany a higher estimated reading grade. They also indicate that adding terminology information to prompts does not necessarily improve all aspects of simplification together [18].

4.5 Ensemble and Multi-Agent Simplification

AIIRLab presented an ensemble combining trained and prompted simplification systems. One branch extracts the main information in a sentence through semantic role labeling, then uses a supervised Ministral 8B model and a version further adapted through DPO. An independent candidate is generated using Gemma 4 31B with few-shot examples. The original sentence, limited document context, and outputs from the three systems are passed to Llama 3.1 70B to produce the final simplification. This final component can rewrite the candidates rather than only select one unchanged. The ensemble achieved sentence-level SARI scores of 46.24 with the original references and 45.08 with the automatic references. Its contribution lies in combining outputs from different simplification approaches. However, the higher SARI score did not correspond to the best FKGL result, and the multiple model calls increase operational cost [5].

Collaborative Document Simplification Using Multi-Agent Systems introduced AgentSimp, which distributes document simplification across agents with different responsibilities. Processing includes preparing document-level guidance and structure, dividing the document into paragraphs, simplifying and reviewing them, explaining terminology and figurative language, and reconstructing and checking the final document. The study used GPT-3.5 and GPT-4 without additional training and compared sequential and parallel communication, as well as direct and incremental reconstruction. Its best Newsela configuration achieved SARI 41.58 and D-SARI 30.72 using GPT-4 with sequential communication and incremental reconstruction. The study demonstrates the potential of assigning complementary simplification roles to multiple agents. However, some review roles did not improve SARI, and API costs limited the evaluation size. Its findings concern document simplification, so the difference in task level must be considered when comparing them with sentence-level results [19].

5 Data Description

5.1 Local development files

The current development data are the Cochrane-auto training, validation, and test sentence files. They are used to train the adapted component, check development decisions, and assess the model locally. The following counts describe the supplied files before project-specific filtering. The three files share the same column schema. [20, 21]

File	Rows	Documents
cochraneauto_sents_train.csv	11,510	849
cochraneauto_sents_val.csv	1,697	119
cochraneauto_sents_test.csv	1,512	117

The complex field supplies the source sentence. The simple field contains its aligned reference text, which is used as a supervised response or an evaluation reference. Operation labels and positional metadata describe the data; their presence does not imply that gold labels are available as model inputs at inference time. [20, 21]

Columns in the development files

Column	Description
pair_id	Document-pair identifier linking the abstract and its plain-language summary.
para_id	Paragraph identifier within the source document.
sent_id	Source sentence identifier within the paragraph.
complex	Original scientific or biomedical sentence.
label	Aligned simplification operation, such as rephrase, split, merge, delete, ignore, or none.
simple	Serialized list of aligned reference simplification segments; an input sentence may align to multiple segments.
simp_sent_id	Identifier or serialized identifiers of aligned simplified segments.
doc_pos	Relative position of the sentence in its document, computed from its one-based position divided by document length.
doc_quint	Document-position quintile, encoded from 1 to 5 in the supplied files.
doc_len	Number of source sentences in the document.

These column descriptions follow the supplied development files and the Cochrane-auto alignment framework. Reference segments must be interpreted together with their alignment, rather than assuming every row has one independent simplified sentence. [20, 21]

5.2 Official Codabench submission input

The previous system processed sentences_src.json and submitted its predictions for Codabench evaluation, obtaining the published final SARI of 45.00. The supplied JSON contains 48,809 source records and six fields. It contains no reference simplification or prediction field. Its total input count is distinct from the subset and reference setting used to calculate a particular platform score. [2, 22]

Column	Description
pair_id	Document or document-pair identifier.
source	Dataset source or evaluation collection.
language	Language code for the source sentence.
para_id	Paragraph identifier.
sent_id	Sentence identifier within the paragraph.
complex	Original sentence to be simplified.

The file combines Cochrane-auto 2026 (3,679 records), Cochrane full 2026 (35,970), Cochrane-auto 2025 (4,293), Cochrane-auto val (1,697), Cochrane-auto test (1,512), Medline (1,010), and SimpleText2024 (648). It includes 20,259 English records and records in ten other languages. The 3,679 Cochrane-auto 2026 records correspond to 175 documents. [22]

The official input also includes the source sentences from the local validation and test files. Accordingly, the complete JSON should not be described as one entirely independent local holdout. The 2026 overview describes a sentence-aligned subset of 32 abstracts and 346 sentences within its evaluation framework. That description does not by itself establish the exact cohort behind the published 45.00 score. Local development evaluation and platform evaluation must therefore be identified separately. [1, 2, 21, 22]

6 Evaluation Metrics

SARI

SARI compares the source, system output, and references to evaluate additions, deletions, and retained content. Higher scores indicate stronger agreement with reference-supported simplification operations. It is the primary project metric, but does not independently verify medical facts or reader comprehension. [2, 23]

BLEU

BLEU measures n-gram overlap with references using modified precision and a brevity penalty. Higher values indicate greater lexical agreement. It is supplementary because a valid simplification can use different wording from the reference. [2, 24]

FKGL and FKGL change

Flesch-Kincaid Grade Level estimates an English reading grade from sentence length and syllable information. Lower values indicate a lower estimated grade. The previous report computes FKGL change as output FKGL minus source FKGL; a negative value indicates improvement in this estimate. It is a surface readability measure, not a direct comprehension test. [9, 25]

Output transformation measures

The previous paper also reports the transformation measures below. They describe changes relative to the source and should be interpreted together. Exact numerical reproduction requires the same preprocessing and scoring implementation. [1, 2]

Measure	Meaning and interpretation
Compression ratio	Output word count divided by source word count in the local report. Values below 1 indicate shorter output; shorter does not necessarily mean more faithful.
Sentence splits	Describes the output sentence count relative to the source. It indicates splitting or combining behavior, not its correctness.
Levenshtein similarity	Normalized similarity based on edit distance. A higher value indicates less surface change.
Exact copies	Proportion of outputs identical to their source under the evaluation normalization. It measures copying, not whether copying was appropriate.
Additions proportion	Describes words introduced relative to the source. It is distinct from the reference-aware addition component of SARI.
Deletions proportion	Describes source words removed from the output. It is distinct from the reference-aware deletion component of SARI.
Lexical complexity	Estimates difficulty of the output vocabulary using the evaluator’s lexical feature. It is a linguistic descriptor, not a medical-accuracy score.

7 Proposed System

7.1 Overall solution

The proposed ensemble contains two independent branches receiving the same original sentence. The first branch uses PEFT, through LoRA or QLoRA, to learn from paired simplification examples. Adaptation occurs before use; candidate generation then runs through inference. The second branch uses the planning-based SPU_DeepSimplify pipeline. [2, 7, 8]

7.2 Possible improvements to the previous system

The existing branch may be refined within its terminology handling, planning, strategy rules, output validation, or selective post-processing. These are possible development directions rather than implemented improvements with established effects. Any refined version should be documented and compared with the published previous system using the same evaluation conditions. The proposed ensemble can then be compared with the reference system and any documented refined version. [2]

7.3 Final ensemble stage

The ensemble component receives the original sentence and the two candidate simplifications. One formulation uses a larger language model without additional task-specific training to select an existing candidate and return it unchanged. Another uses the model to generate a new simplification informed by both candidates and grounded in the original sentence. The final choice remains open. The reviewed candidate-selection and fusion systems motivate these alternatives, but do not establish their performance for the proposed combination. [3, 5]

Figure 1 summarizes the proposed inference flow. The source is passed to the final stage as well as to both branches. The diagram describes the architecture and does not report a completed experiment or an achieved ensemble result.

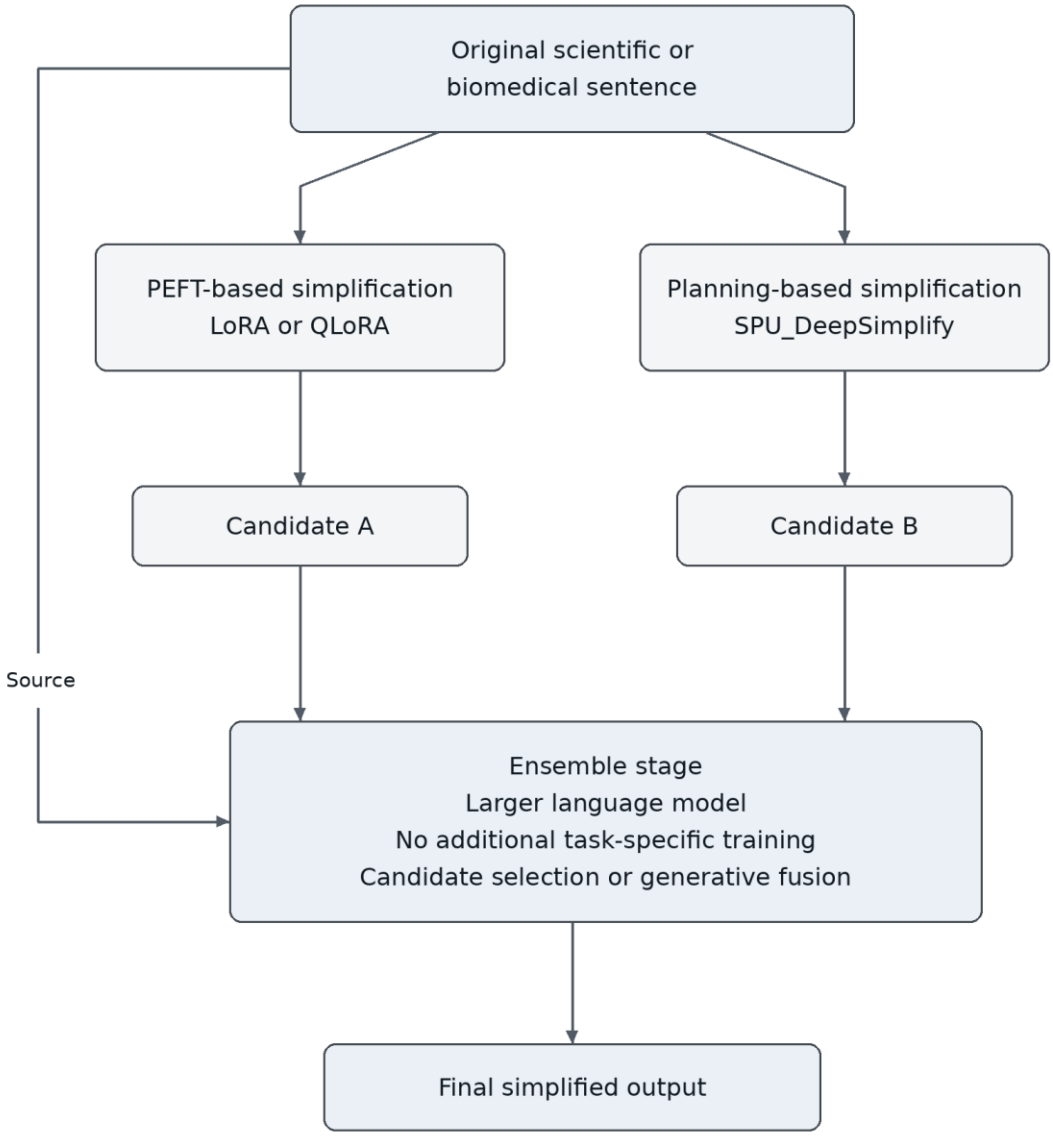


Figure 1 Proposed architecture

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